Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/01/2016
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	Neurobehavioral and CNS-Related Physiological Changes Following Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	catherine.davis-takacs@usuhs.edu	**Fax:**	FY 301-400-4023
PI Organization Type:	NON-PROFIT	**Phone:**	301-400-4596
Organization Name:	Henry M. Jackson Foundation		
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City:	Bethesda	**State:**	MD
Zip Code:	20817-1891	**Congressional District:**	8
Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	01/01/2015	**End Date:**	12/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	NNX15AC71G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Assessing the biological consequences of living in the space radiation environment represents one of the highest priority areas of NASA research. Of critical importance is the need for an assessment of the vulnerabilities of the central nervous system (CNS) leading to functional neurobehavioral changes during long-term space missions, and the development of effective countermeasures to such risks. The present proposal addresses this need via the application of an animal model to: 1) monitor indices of circadian function (i.e., spontaneous locomotor activity, body temperature) over various time points following irradiation; 2) determine the degree to which changes in circadian function are correlated with the severity of behavioral deficits in a rat psychomotor vigilance task; and 3) examine the underlying neurochemical alterations that accompany these radiation-induced changes in circadian function and neurobehavioral performance. Prior research has 1) identified rats that are sensitive to radiation-induced deficits in neurobehavioral function; 2) identified changes in the dopamine neurotransmitter systems of these radiation-sensitive rats; and 3) identified an increased inflammatory response in the brain from 1-14 days post-irradiation. The current proposal will determine how the immediate effects of irradiation impact subsequent neurobehavioral deficits by assessing physiological markers of circadian function that are sensitive to changes in immunological functions and the onset of pathophysiological states. Groups of animals with implanted biotelemetry transmitters will be trained on a rodent version of the human Psychomotor Vigilance Test (the rPVT) and exposed to radiation; fluctuations in spontaneous locomotor activity and body temperatures will be assessed immediately following exposure and tracked over time and compared to the neurobehavioral performances. Individual variations in these physiological responses following radiation will be assessed for use as possible biomarkers for radiation sensitivity resulting in neurobehavioral deficits. Likely mechanisms of damage to the CNS following radiation exposure will be examined using brain tissue, in addition to an examination of cytokine levels in blood and peripheral organ systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this work will provide knowledge regarding the effects of radiation on neurobehavioral function and circadian fluctuations in core body temperature and locomotor activity. This information will enhance our understanding of radiation's effects and can be used to determine the effects of terrestrial radiation exposures and ways to mitigate their adverse effects.
Task Progress:	One cohort of 40 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to Brookhaven National Laboratory (BNL) in November 2015. These rats were sham exposed or exposed to whole-body protons (150 MeV/n) at 25 and 100 cGy. in addition to a third group exposed to 100 cGy head-only protons. Telemetry data was recorded at BNL prior to the radiation exposure and for several days after the exposure until the rats were shipped back to Johns Hopkins University (JHU) for follow-up behavioral testing. Telemetry data was acquired 24-hrs per day for ~20 weeks post-proton exposure. During this time, rats also performed the rPVT five days per week in 30-min sessions. Similar to our previous studies and like head-only proton exposure, whole-body proton exposure resulted in significant rPVT deficits in a subset of rats, which included decreased accuracy and increased premature responding, lapses in attention, and response latencies (median and mean reaction times). Analyses are on-going to determine if all irradiated rats displayed normal rhymicity in body temperature and activity levels. Additional analyses of gene expression in the liver and protein expression in the brain are on going. A second series of experiments is on going to determine if oxygen exposure results in significant impairments in sustained attention, as measured by the rodent psychomotor vigilance test (rPVT), recognition memory, as measured by the social odor recognition memory test, and indices of circadian rhythmicity of activity and body temperature. A cohort of 40 rats was trained on the rPVT, implanted with biotelemetry devices, and shipped to BNL in June 2016. These rats were sham exposed or exposed to whole-body oxygen ions (16O, 1000 MeV/n) at 5 cGy or 25 cGy. While behavioral data analysis is on going, thus far rats display significant deficits in recognition memory at 30-days post-exposure. Interestingly, these rats do not display deficits in novelty recognition, but are unable to retain the memory of a novel social odor over a 24-hr retention interval. These deficits were similar between 5 and 25 cGy. Individual oxygen-exposed rats at both 5 and 25 cGy displayed significant increases in impulsive responding on the rPVT, primarily at the longer wait times, suggesting that oxygen exposure results in an inability to wait a longer time for the onset of a stimulus. These deficits are not related to motor impairments, given that these irradiated animals continued to make responses on the rPVT, but these responses were prior to the onset of the light cue. Measures of rhythmicity are also being analyzed in this cohort of rats.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Gupta SR, Hienz RD, Davis CM. "Psychostimulants differentially alter performance on the rodent psychomotor vigilance test following radiation exposure." Presented at Experimental Biology 2016, San Diego, CA, April 2-6, 2016. FASEB Journal. 2016 Apr;30(1 Suppl):1184.9. http://www.fasebj.org/content/30/1_Supplement.toc for searching. , Apr-2016
Articles in Peer-reviewed Journals	Davis CM, Roma PG, Hienz RD. "The rodent psychomotor vigilance test (rPVT): A method for assessing neurobehavioral performance in rats and mice." Journal of Visualized Experiments, in press, as of November 2016. , Nov-2016
Articles in Peer-reviewed Journals	Davis CM, DeCicco-Skinner KL, Hienz RD. "Deficits in sustained attention and changes in dopaminergic protein levels following exposure to proton radiation are related to basal dopaminergic function." PLoS One. 2015 Dec 10;10(12):e0144556. http://dx.doi.org/10.1371/journal.pone.0144556 ; PubMed PMID: 26658810; PubMed Central PMCID: PMC4684339 , Dec-2015
Articles in Peer-reviewed Journals	Davis CM, Roma PG, Hienz RD. "A rodent model of the human psychomotor vigilance test: Performance comparisons." J Neurosci Methods. 2016 Feb 1;259:57-71. http://dx.doi.org/10.1016/j.jneumeth.2015.11.014 ; PubMed PMID: 26639896 , Feb-2016

Awards	King SE, Davis CM (Mentor). "Summer Undergraduate Research Internship (Stipend) awarded by the American Society for Pharmacology and Experimental Therapeutics. Supported 10-week summer research internship in the laboratory. May - July 2016." May-2016